Experiment 'p_5_9_f_cone_surface' Results

June 6, 2026

Experiment outcome: FAILURE_VERIFY
Bad responses: 1
Responses containing `assume` : 0
Responses containing `expect` : 0
Resolution attempts: 3
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 3

Problem Specification

Problem name: p_5_9_f_cone_surface
Natural language statement: Write a method that computes the surface area of a cone with a circular base with a residual error less than 0.01, given radius and height.
Method signature: p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)

Ensures

- `-0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) + (r * r)) < 0.01`

Requires

- `r > 0.0`

GenAI interactions

Below you will find all interactions between the 'user' (program) and the 'assistant' (OpenAI).

Program → GenAI

```
You are given the following task to perform in Dafny:

Write a method that computes the surface area of a cone with a circular base with a
    ↪ residual error less than 0.01, given radius and height.

The signature should be:

p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)

The method should respect the following contract:

requires r > 0.0, ensures -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159
    ↪  * r)) - r)) - ((h * h) + (r * r)) < 0.01

Produce and show only the Dafny body of this method, including the curly braces that
    ↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
    ↪  'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnjeqsqtSOBAttJh7Tx3qTzxDtET8
Model: gpt-5.5-2026-04-23
Created at: 1780745952
Finish reason: stop
Usage: 406 tokens in, and 4299 tokens out

```
{
  var s: real :| s * s == h * h + r * r;
  surface := 3.14159 * r * (r + s);

  assert 3.14159 * r != 0.0;
  assert surface / (3.14159 * r) == r + s;
  assert (surface / (3.14159 * r)) - r == s;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
      ↪  (r * r)) == 0.0;
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_5_9_f_cone_surface/program.dfy(5,14): Error: cannot establish the
    ↪ existence of LHS values that satisfy the such-that predicate. Note, no trigger was
    ↪  found for the such-that predicate, which may be the reason the proof failed. To
```

```
    ↪ give a trigger explicitly, use the {:trigger} attribute. For more information, see
    ↪  the section on quantifier instantiation rules in the reference manual.
  |
5 | var s: real :| s * s == h * h + r * r;
  | ^^^^^^^^^^^^^^^^^^^^^^^^^^


Dafny program verifier finished with 1 verified, 1 error

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-DnjgkbpOegzufzVmS6R0VyAFoPZ0e
Model: gpt-5.5-2026-04-23
Created at: 1780746070
Finish reason: stop
Usage: 791 tokens in, and 10345 tokens out

```
{
  var y := h * h + r * r;
  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert y > 0.0;

  var low := 0.0;
  var high := 1.0;

  while high * high < y
    invariant low == 0.0
    invariant low <= high
    invariant high > 0.0
    invariant low * low <= y
    decreases y - high * high
  {
    var oldHigh := high;
    assert oldHigh > 0.0;
    assert oldHigh * oldHigh < (2.0 * oldHigh) * (2.0 * oldHigh);
    high := 2.0 * oldHigh;
    assert y - high * high < y - oldHigh * oldHigh;
  }

  assert low * low <= y;
  assert y <= high * high;

  while high * high - low * low >= 0.01
    invariant 0.0 <= low
    invariant low <= high
```

```
    invariant high > 0.0
    invariant low * low <= y
    invariant y <= high * high
    decreases high * high - low * low
  {
    var oldLow := low;
    var oldHigh := high;
    var mid := (oldLow + oldHigh) / 2.0;

    assert oldHigh * oldHigh - oldLow * oldLow >= 0.01;
    assert oldLow < oldHigh;
    assert mid - oldLow == (oldHigh - oldLow) / 2.0;
    assert oldHigh - mid == (oldHigh - oldLow) / 2.0;
    assert oldLow < mid;
    assert mid < oldHigh;
    assert mid > 0.0;

    if mid * mid <= y {
      assert mid * mid - oldLow * oldLow == (mid - oldLow) * (mid + oldLow);
      assert mid * mid - oldLow * oldLow > 0.0;
      low := mid;
      assert oldHigh * oldHigh - low * low < oldHigh * oldHigh - oldLow * oldLow;
    } else {
      assert oldHigh * oldHigh - mid * mid == (oldHigh - mid) * (oldHigh + mid);
      assert oldHigh * oldHigh - mid * mid > 0.0;
      high := mid;
      assert high * high - oldLow * oldLow < oldHigh * oldHigh - oldLow * oldLow;
    }
  }

  assert high * high - low * low < 0.01;
  assert high * high - y >= 0.0;
  assert high * high - y <= high * high - low * low;
  assert high * high - y < 0.01;

  surface := 3.14159 * r * (r + high);

  assert 3.14159 > 0.0;
  assert 3.14159 * r > 0.0;
  assert 3.14159 * r != 0.0;
  assert surface == (3.14159 * r) * (r + high);
  assert surface / (3.14159 * r) == r + high;
  assert (surface / (3.14159 * r)) - r == high;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
      ↪  (r * r)) == high * high - y;
  assert -0.01 < high * high - y;
  assert high * high - y < 0.01;
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_5_9_f_cone_surface/program.dfy(8,2): Error: assertion might not hold
  |
8 | assert y > 0.0;
  | ^^^^^^

problem_data/p_5_9_f_cone_surface/program.dfy(30,2): Error: decreases expression might
    ↪ not decrease
   |
30 | while high * high - low * low >= 0.01
   | ^^^^^

problem_data/p_5_9_f_cone_surface/program.dfy(74,2): Error: assertion might not hold
   |
74 | assert surface / (3.14159 * r) == r + high;
   | ^^^^^^


Dafny program verifier finished with 1 verified, 3 errors

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dnjkiex2veAsp8NcJ23noYmZHrHfr
Model: gpt-5.5-2026-04-23
Created at: 1780746316
Finish reason: stop
Usage: 1928 tokens in, and 6498 tokens out

```
{
  var y := h * h + r * r;

  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert h * h + r * r >= r * r;
  assert h * h + r * r > 0.0;
  assert y > 0.0;
  assert y >= 0.0;

  var low := 0.0;
  var high := y + 1.0;

  assert low * low <= y;
  assert high > 0.0;
  assert low <= high;
  assert y * y >= 0.0;
```

```
assert (y + 1.0) * (y + 1.0) - y == y * y + y + 1.0;
assert (y + 1.0) * (y + 1.0) >= y;
assert y <= high * high;

while high * high - low * low >= 0.01
  invariant 0.0 <= low
  invariant low <= high
  invariant high > 0.0
  invariant low * low <= y
  invariant y <= high * high
  decreases high - low
{
  var oldLow := low;
  var oldHigh := high;
  var mid := (oldLow + oldHigh) / 2.0;

  assert oldHigh * oldHigh - oldLow * oldLow >= 0.01;
  assert oldHigh * oldHigh - oldLow * oldLow > 0.0;
  if oldLow == oldHigh {
    assert oldHigh * oldHigh - oldLow * oldLow == 0.0;
    assert false;
  }
  assert oldLow < oldHigh;
  assert mid == oldLow + (oldHigh - oldLow) / 2.0;
  assert oldLow < mid;
  assert mid < oldHigh;
  assert mid > 0.0;

  if mid * mid <= y {
    low := mid;
    assert low * low <= y;
    assert y <= high * high;
  } else {
    high := mid;
    assert y < high * high;
    assert y <= high * high;
    assert low * low <= y;
  }

  assert high - low == (oldHigh - oldLow) / 2.0;
  assert oldHigh - oldLow > 0.0;
  assert (oldHigh - oldLow) / 2.0 < oldHigh - oldLow;
  assert high - low < oldHigh - oldLow;
}

assert high * high - low * low < 0.01;
assert high * high - y >= 0.0;
assert high * high - y <= high * high - low * low;
assert high * high - y < 0.01;

var denom := 3.14159 * r;
```

```
  var z := r + high;
  surface := denom * z;

  assert 3.14159 > 0.0;
  assert denom > 0.0;
  assert denom != 0.0;
  assert denom / denom == 1.0;
  assert (denom * z) / denom == (denom / denom) * z;
  assert (denom * z) / denom == z;
  assert surface / denom == z;
  assert denom == 3.14159 * r;
  assert surface / (3.14159 * r) == z;
  assert (surface / (3.14159 * r)) - r == high;
  assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
    ↪  (r * r)) == high * high - y;
  assert -0.01 < high * high - y;
  assert high * high - y < 0.01;
}
```

Program → GenAI

```
When using dafny verify, the below error is emitted and verify fails:

problem_data/p_5_9_f_cone_surface/program.dfy(11,2): Error: assertion might not hold
   |
11 | assert y > 0.0;
   | ^^^^^^

problem_data/p_5_9_f_cone_surface/program.dfy(25,2): Error: decreases expression might
    ↪ not decrease
   |
25 | while high * high - low * low >= 0.01
   | ^^^^^

problem_data/p_5_9_f_cone_surface/program.dfy(78,2): Error: assertion might not hold
   |
78 | assert denom / denom == 1.0;
   | ^^^^^^

problem_data/p_5_9_f_cone_surface/program.dfy(79,2): Error: assertion might not hold
   |
79 | assert (denom * z) / denom == (denom / denom) * z;
   | ^^^^^^


Dafny program verifier finished with 1 verified, 4 errors

Correct the error in code by altering only the method body. Produce and show only the
    ↪ Dafny body, including the curly braces that surround it. Do not show the signature
    ↪  nor contract. You must not use 'assume' nor 'expect' anywhere in your code.
```

GenAI → Program

System fingerprint: null
ID: chatcmpl-Dnjn6aGMhUMtBkPfv8YkSjH5SYRFG
Model: gpt-5.5-2026-04-23
Created at: 1780746464
Finish reason: length
Usage: 3193 tokens in, and 5733 tokens out
The below is a raw response from GenAI that could not be parsed. It is likely that GenAI got 'stuck' in a loop and hit the pre-set token limit (i.e., finish reason is 'length').

Final Program

```
method p_5_9_f_cone_surface(r: real, h: real) returns (surface: real)
       requires r > 0.0
       ensures -0.01 < (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)
           ↪ ) - ((h * h) + (r * r)) < 0.01
{
  var y := h * h + r * r;

  assert h * h >= 0.0;
  assert r * r > 0.0;
  assert h * h + r * r >= r * r;
  assert h * h + r * r > 0.0;
  assert y > 0.0;
  assert y >= 0.0;

  var low := 0.0;
  var high := y + 1.0;

  assert low * low <= y;
  assert high > 0.0;
  assert low <= high;
  assert y * y >= 0.0;
  assert (y + 1.0) * (y + 1.0) - y == y * y + y + 1.0;
  assert (y + 1.0) * (y + 1.0) >= y;
  assert y <= high * high;

  while high * high - low * low >= 0.01
    invariant 0.0 <= low
    invariant low <= high
    invariant high > 0.0
    invariant low * low <= y
    invariant y <= high * high
    decreases high - low
  {
    var oldLow := low;
    var oldHigh := high;
    var mid := (oldLow + oldHigh) / 2.0;
```

```
  assert oldHigh * oldHigh - oldLow * oldLow >= 0.01;
  assert oldHigh * oldHigh - oldLow * oldLow > 0.0;
  if oldLow == oldHigh {
    assert oldHigh * oldHigh - oldLow * oldLow == 0.0;
    assert false;
  }
  assert oldLow < oldHigh;
  assert mid == oldLow + (oldHigh - oldLow) / 2.0;
  assert oldLow < mid;
  assert mid < oldHigh;
  assert mid > 0.0;

  if mid * mid <= y {
    low := mid;
    assert low * low <= y;
    assert y <= high * high;
  } else {
    high := mid;
    assert y < high * high;
    assert y <= high * high;
    assert low * low <= y;
  }

  assert high - low == (oldHigh - oldLow) / 2.0;
  assert oldHigh - oldLow > 0.0;
  assert (oldHigh - oldLow) / 2.0 < oldHigh - oldLow;
  assert high - low < oldHigh - oldLow;
}

assert high * high - low * low < 0.01;
assert high * high - y >= 0.0;
assert high * high - y <= high * high - low * low;
assert high * high - y < 0.01;

var denom := 3.14159 * r;
var z := r + high;
surface := denom * z;

assert 3.14159 > 0.0;
assert denom > 0.0;
assert denom != 0.0;
assert denom / denom == 1.0;
assert (denom * z) / denom == (denom / denom) * z;
assert (denom * z) / denom == z;
assert surface / denom == z;
assert denom == 3.14159 * r;
assert surface / (3.14159 * r) == z;
assert (surface / (3.14159 * r)) - r == high;
assert (((surface / (3.14159 * r)) - r) * ((surface / (3.14159 * r)) - r)) - ((h * h) +
    ↪  (r * r)) == high * high - y;
assert -0.01 < high * high - y;
```

```
  assert high * high - y < 0.01;
}
```

Total Token Usage

Input tokens: 6318
Output tokens: 26875
Reasoning tokens: 24699
Sum of 'total tokens': 33193

Experiment Timings

Overall Experiment started at 1780745951909, ended at 1780746604058, lasting 652149ms (652.15 seconds)
Iteration #4 started at 1780746464246, ended at 1780746604058, lasting 139812ms (139.81 seconds)
Iteration #1 started at 1780745951912, ended at 1780746070548, lasting 118636ms (118.64 seconds)
Iteration #2 started at 1780746070549, ended at 1780746315725, lasting 245176ms (245.18 seconds)
Iteration #3 started at 1780746315725, ended at 1780746464245, lasting 148520ms (148.52 seconds)